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Comparing Large Language Model Methodologies for Life Cycle Inventory Data Extraction

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Abstract

Life Cycle Inventory (LCI) data is essential for the evaluation of the environmental impacts of resource- and energy-intensive industrial activities, including the extraction and processing of materials for energy storage systems, such as lithium-ion (LIB) or sodium-ion batteries (SIB). However, the process of extracting relevant data from a variety of technical documentation and literature, including mining reports, is often time-consuming. The challenges encountered in this process include semi-structured content, inconsistent formatting, and domain-specific terminology. Recent advances in natural language processing (NLP) and large language models (LLMs) offer promising tools to address these challenges by enabling automated understanding and extraction of relevant information from complex documentation. This study compares two advanced NLP approaches: Quantized Low-Rank Adaptation (QLoRA)-based question answering and Retrieval-Augmented Generation (RAG). In both approaches, the open-source LLM model LLaMA is employed and used for the extraction of LCI data. The QLoRA-based method employs fine-tuning to adapt the model to perform a specific task. By contrast the RAG approach uses a LLaMA model alongside a system that identifies and retrieves relevant information from a document section as context for the answer generation. The evaluation is conducted on a manually annotated dataset comprising technical reports and documentation related to LIB and SIB. The results indicate that the fine-tuned model performs well in accurately identifying relevant LCI information. However, the RAG approach is unable to capture all necessary LCI details, despite its ability to filter out irrelevant data. This comparative study provides insights into the applicability of LLM-based methods for LCI data extraction from technical and environmental documentation. Hence, demonstrating the potential of AI-supported inventory data acquisition in the domain of energy storage technologies and thereby contributing to the development of more efficient approaches for conducting life cycle assessment.

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Keywords: life cycle inventory;large language model; natural language processing; retrieval-augmented generation; quantized low rank adaption, battery

1. Introduction

Energy storage systems play a central role in supporting decarbonized mobility and renewable energy integration [1]. In the European Union, the European Green Deal and the revised Battery Regulation (EU 2023/1542) emphasize sustainability and circularity throughout the battery life cycle, including due diligence, traceability, and value chain transparency [2,3]. In this context, various battery technologies are currently in focus: conventional lithium-ion batteries (LIBs) dominate the field at present, while next-generation alternatives, such as sodium-ion batteries (SIBs), are emerging as potential contenders. The main difference lies in the cathode materials: LIBs rely on lithium, cobalt and nickel, all critical resources, whereas SIBs use

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sodium and iron, which are abundant and potentially more cost-effective [4,5]. In addition to technical performance, sustainability has become a crucial factor for both types of batteries in terms of policy and societal considerations. A well-established method to assess sustainability is life cycle assessment (LCA) which quantifies environmental impacts of products, processes and services. It comprises four phases: goal and scope definition, inventory analysis, impact assessment, and interpretation [6]. LCA relies mainly on life cycle inventory (LCI) data, capturing the quantification of material and energy flows [7]. However, acquiring high-quality LCI data, particularly primary data, remains challenging due to the fragmented and heterogenous nature of sources, such as reports and process documentation [8–11]. While LIBs benefit from their high technology readiness level (TRL) and the availability of primary data on an industrial scale, such as technical reports and test data, LCI data for SIBs is often based on laboratory information or existing literature due to their low TRL as an emerging technology. In the absence of primary data, LCAs often use prospective approaches, which involve a systematical evaluation of future developments and their potential impacts. This enables informed decision-making regarding the technology's viability, environmental sustainability and possible future directions. To automate the extraction of the LCI-relevant information, recent research has explored the use of artificial intelligence (AI) and natural language processing (NLP) [12,13].

Although case studies suggest that large language models (LLMs) could improve efficiency, the use of NLP in LCA workflows is still in its infancy [14]. So far, few studies have compared architectures, such as fine-tuning and retrieval augmented generation (RAG) in the context of battery-related documentation. This is crucial for optimizing data processing and enhancing the efficiency of LCAs in this rapidly evolving field. This study aims to address this gap by systematically evaluating NLP-based approaches for extracting LCI data from LIB and SIB documentation.

2. Recent advances in LCI data processing using LLMs

LCA practitioners often encounter significant challenges related to data accessibility and organization. Relevant LCI information is frequently scattered across semi-structured reports and technical documents which using inconsistent terminology and complex language [8–10,15]. This dispersion makes time-consuming manual data collection and standardization, requiring expert knowledge for cross-referencing and formatting [14,15]. These challenges cause delays in LCA workflows, especially for emerging technologies with fragmented documentation, such as in the battery industry, where access to consistent and comprehensive data is essential.

2.1 Recent large language model approaches

Recent advances in NLP and LLMs offer promising solutions for automated data extraction [12,13]. Pre-trained transformer models such as BERT, GPT and LLaMA perform well in structuring technical content [16,17]. Open-source LLaMA variants support diverse tasks, including question answering and summarization [18], and newer versions (LLaMA2, LLaMA3) enhance stability, document handling, and processing speed [16,19,20]. These advancements improve accuracy and efficiency in analyzing large technical datasets, potentially streamlining LCA workflows.

Fine-tuning involves adapting LLMs for specific domains by updating their parameters using targeted data. This improves specialization but requires substantial computational resources [21]. For instance, a model initially trained on general text data can be fine-tuned using a dataset focused on environmental regulations to improve its ability to respond accurately to queries related to LCAs. Techniques such as Low-Rank Adaptation (LoRA) and Quantized LoRA (QLoRA) reduce costs by introducing trainable adapters or quantization while maintaining strong accuracy [21–23]. For instance, QLoRA might allow practitioners to deploy fine-tuned models on less powerful hardware, making advanced LLM capabilities more accessible without requiring extensive computational power. This combination of methods enables practitioners to deploy fine-tuned models efficiently, tailoring them to domain-specific needs under limited resources.

RAG instead links LLMs with external sources, such as technical documentation, in order to retrieve relevant data. This information is then added to the prompt and used by the model to enhance its responses. For example, if a LCA practitioner queries the model about particular environmental impacts of LIB, RAG retrieves the most recent data from suitable technical reports or LCIs. This would provide users with contemporary and precise information that surpass static pre-training. This approach could enable LCA practitioners to efficiently access domain-specific or up-to-date information, making it valuable for conducting comprehensive LCAs [24].

2.2 Current application of LLM techniques in LCI

Although AI applications in LCA / LCI are still emerging, several studies highlight LLMs' strong potential for automating data collection through fine-tuning [14,25,26]. Case studies demonstrate improved data acquisition and traceability, particularly for multilingual or semi-structured technical and regulatory sources [12,27–29]. Building on this, recent work has integrated LLMs with retrieval frameworks and structured knowledge systems. For example, Wang et al. [12] propose a RAG-based architecture that can extract carbon footprint data in real time from technical databases, thereby improving efficiency and coverage. Greif et al. [27] use LLMs with knowledge graphs to estimate missing LCI values (e.g. energy use or CO_2 emissions), which are useful in the early stages of design. Furthermore, Tai et al. [28] and Turhan [29] demonstrate that NLP and fine-tuned LLMs extract sustainability indicators from stakeholder reports and datasets. The existing research thus explores both fine-tuning and RAG individually, yet few studies directly compare their effectiveness for domain-specific LCI extraction, particularly for batteries.

Despite these advances, the comparative advantages of fine-tuning and RAG for battery-related LCI tasks remain unclear. The existing studies assess each method individually, yet a systematic evaluation of their performance under realistic conditions remains limited. Therefore, this study addresses it by directly comparing fine-tuning and RAG for automated LCI data extraction in the battery domain, aiming to reduce manual effort while improving data consistency and traceability.

3. NLP-based approach for LCI data extraction

The rapid advancement of NLP presents exciting opportunities for extracting LCI data. The combination of LLMs with retrieval systems improves access to information from technical sources [12]. Literature also suggests that domain-specific fine-tuning could automate the data collection process even further [14]. Building on recent developments and their potentials, this study compares two LLM-based approaches - QLoRA-based fine-tuning and RAG - for extracting LCI-relevant data from battery-related documents, in which essential information is often dispersed across various formats and sources. Both approaches rely on the open-source LLaMA model, which was selected due to its demonstrated performance in technical NLP tasks and its efficient adaptability through parameter-efficient fine-tuning methods such as LoRA [17,18]. In contrast to proprietary models such as ChatGPT, LLaMA offers full transparency. Open-source availability, and the possibility of local deployment without dependence on commercial application program interfaces (APIs) or cloud service. These features ensure reproducibility and data privacy, which are essential for research-oriented and domain-specific applications such as LCI extraction.

3.1 Document selection, data annotation and preprocessing

In order to evaluate the two methods, a technical report containing relevant information regarding LCI modelling is chosen. The document includes data on resource content, quantities and energy consumption, all of which are important for understanding the environmental impacts of LIBs [8–11,30]. To enable a robust evaluation of the extraction performance, the document is annotated manually by a domain expert and information relevant to LCI, such as energy usage and chemical composition, is recorded in a structured format. These annotations form the basis of the training data for QLoRA-based fine-tuning and provide a reference benchmark for evaluating both approaches. In addition, a LCA study on SIBs is selected which provides detailed LCI data on various lab-scale synthesis routes for cathode materials [31]. This makes it a suitable document for assessing the transferability of the methodology to prospective assessments of emerging technologies.

3.2 QLoRA-based fine-tuning approach

To create the fine-tuning data, ChatGPT API is used to generate question-and-answering pairs from annotated datapoints. Positive examples (80%) include extractable datapoints while 20% of negative examples contain unanswerable yet realistic questions, thereby improving robustness of the model's responses and reducing hallucinations [32]. All datasets undergo manual quality control before being split into training (80%), validation (10%), and test (10%) datasets in JSON format. Fine-tuning is performed using efficient QLoRA, which only modifies key weight modules in order to reduce the resource demand. Training runs for five epochs with small batch size of four and learning rate adjustments.

Performance is monitored via training loss, evaluation loss and perplexity. Training loss indicates how well the model aligns with the training data. For example, a decreasing training loss suggests that the model effectively learns from the provided examples. In contrast, evaluation loss measures performance on unseen data. High evaluation loss alongside low training loss may cause signal overfitting, suggesting that the model is struggling with new data. Perplexity quantifies uncertainty in predicting the next word: lower scores reflect better predictive accuracy, such as successfully completing a phrase compared to erratic outputs. The training process is monitored in terms of steps, representing parameter updates, and epochs, indicating how many times the dataset has been processed. These metrics enable the analysis of convergence, stability and generalizability to new data.

3.3 Retrieval augmented generation (RAG)

In contrast, RAG leaves the model's parameters unaltered. Instead, it improves the model's responses by injecting high-quality context from retrieved documents. The document is segmented into text sections of up to 1,200 characters. These segments are transformed into dense semantic embeddings using vectorization and indexed using FAISS (Facebook AI Similarity Search), enabling efficient and scalable searches across the document. Furthermore, LangChain is used which facilitates persistent storage and integration with LLMs. For example, when the user asks a question, the system retrieves the most relevant text segment from the FAISS index, prioritizing diversity in the responses. During inference, relevant text is retrieved and appended to the user query before being passed to the LLM for answer generation. Thus, the RAG process effectively combines semantic searching with context-aware answer generation, minimizing the likelihood of incorrect information, which is known as hallucinations. This ensures that all answers can traced back to their original sources.

3.4 Evaluation methods

To evaluate the two approaches, the generated test dataset is applied and the correct answers are documented alongside the predictions. The results are saved automatically, enabling both quantitative and qualitative evaluations. This ensures transparency and traceability, which are important in case where auditability and evidence citation are critical. Performance is assessed using standard metrics in information retrieval and text classification: accuracy, precision, recall, and F1-score. An accuracy value of >0.9 indicates high reliability, while precision reduces false positives and recall minimizes false negatives. The F1-score captures the trade-offs between the two, with values greater than 0.8 being regarded as good. Confusion matrices further support error analysis and the identification of systematic weaknesses. The formulas and thresholds for the metrics are provided in Table 1.

Table 1: Metrics calculation and typical thresholds

Metric formular	Threshold
$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$	> 0.9
$Precision = \frac{TP}{TP + FP}$	> 0.8
$Recall = \frac{TP}{TP + FN}$	> 0.8
$F1 - Score = 2\frac{Precison\ Recall}{Precision + Recall}$	> 0.8

TP: true positive, TN = true negatives, FP: false positive, FN: false negative

4. Case Study on documentation for lithium mining

For battery technologies and their value chain, the LCI requires reliable data, for example from the extraction of raw material such as lithium. Lithium is essential for producing LIBs, especially for electrical vehicles and renewable energy storage systems, and is therefore a key part of the transition to sustainable energy [1,33]. This study uses the technical report from the Greenbushes Lithium Operations in Western Australia, prepared for Talison Minerals Pty Ltd and Talison Lithium Ltd of 2009 [30], as primary data source. This report is significant because it offers details and provides fundamental operational data. Specifically, it covers the extraction of raw materials and the production processes in lithium mining and can be used directly or serves as the basis for calculating specific inventory flows [9]. It also includes insights into exploration and drilling activities, along with various analyses, operational requirements and applied methodologies.

4.1 QLoRA-based fine-tuning approach

To improve the model's performance, a domain expert annotated the technical report on lithium mining in Australia, resulting in 156 datapoints across ten chapters. The data points are then transferred into clear sentences to generate question-and-answer pairs with the API LLM. This results in a total of 1,627 positive pairs, representing 80% of the total datasets. To ensure the model learns effectively, negative datasets are also created using sections of the reports lacking relevant data points, resulting in 328 question-and-answer pairs. In total, 1,955 data sets are compiled and randomly divided into training (80%), validation (10%) and test data (10%) sets. The datasets are reviewed manually to ensure consistency and quality of the training data. Furthermore, this approach ensures that the model accurately reflects the complexities of the battery domain. For fine-tuning, the LLaMA-3.1-8B-Instruct model with specified parameters, stated in 3.3.1, is used. During the training, which updates the model weights, the metrics, such as loss and perplexity are logged to monitor the performance and indicate the model's behavior on unseen data. For example, given the input "The subtotal for reserves includes 8.24 Mt of ore … equating to an overall strip ratio of 2.3," an untrained model might output "The reserves contain ore and waste material" instead of the gold label "The strip ratio refers to the waste-to-ore ratio, which in this case is 2.3.".

Figure 1 shows how training and evaluation loss change over five epochs. Both losses decrease sharply during the initial training phase, after which they stabilize. This indicates that the model adapts quickly to the task before achieving stable performance. Figure 2 depicts the corresponding evaluation perplexity, which falls in line with the evaluation loss, stabilizing at a low level that reflects improved predictive accuracy on unseen data (validation data). The close alignment of the training and evaluation curves suggests that there is no substantial overfitting during fine-tuning. For the final investigation, the fine-tuned model is applied to generate answers from the reserved test dataset. These answers are then compared to the reference output using a string similarity measure, which provides a direct and interpretable indication of textual agreement. These metric suits short, factual LCI statements, where precise terminology ensures semantic accuracy. Predictions with a similarity score above a predefined threshold are classified as correct, while borderline cases underwent manual review for accuracy.

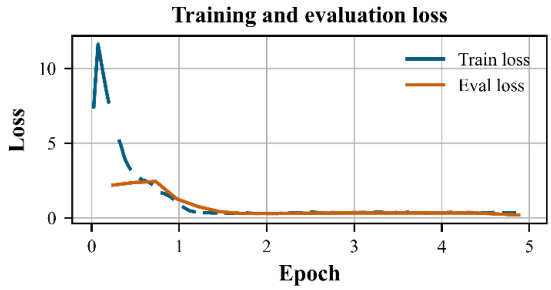


Figure 2: Training and evaluation loss curve for QLoRA based fine-tuning approach

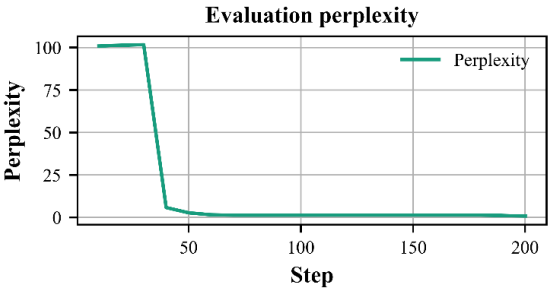


Figure 1: Perplexity curve for QLoRA-based fine-tuning approach

4.2 Retrieval augmented generation

For the document-based retrieval, the chapter-wise text files are divided into smaller, overlapping sections using a method called *Recursive Character Text Splitter*. Each 1,200-character chunk, with an overlap of 200 characters, is organized hierarchically by paragraph and sentence. This method preserves the context and meaning of the section, which is crucial when dealing with complex technical information about batteries, especially lithium mining. Each chunk is annotated with metadata for traceability purposes and stored in a JSON file. Resulting in 150 chunks that formed the basis for the semantic retrieval process. To prepare for retrieval, the processed text chunks are transformed using a light weight sentence transformer model to help with grouping similar text segments together. These embeddings are then indexed using FAISS to facilitate efficient searches. When a user poses a question, the system retrieves the six most relevant chunks based on semantic similarity ensuring that the model generates contextually relevant answers about lithium mining. RAG evaluation is conducted with the reserved test dataset, inserting selected chunks into the LLaMA model's prompt. The model produces concise answers based on the retrieved context, with specific formatting for numeric outputs. Predictions are evaluated using semantic and character-level similarity thresholds, and borderline cases are flagged for manual review.

4.3 Quantitative evaluation metrics

In addition to evaluating the fine-tuned model and RAG setup, the performance of the original LLaMA-3.1-8B-Instruct model (basic model) is analyzed. To assess the predictions comprehensively, the quantitative metrics such as accuracy, precision, recall and F1-score are used. The evaluation is based in predefined thresholds, providing a clear framework for

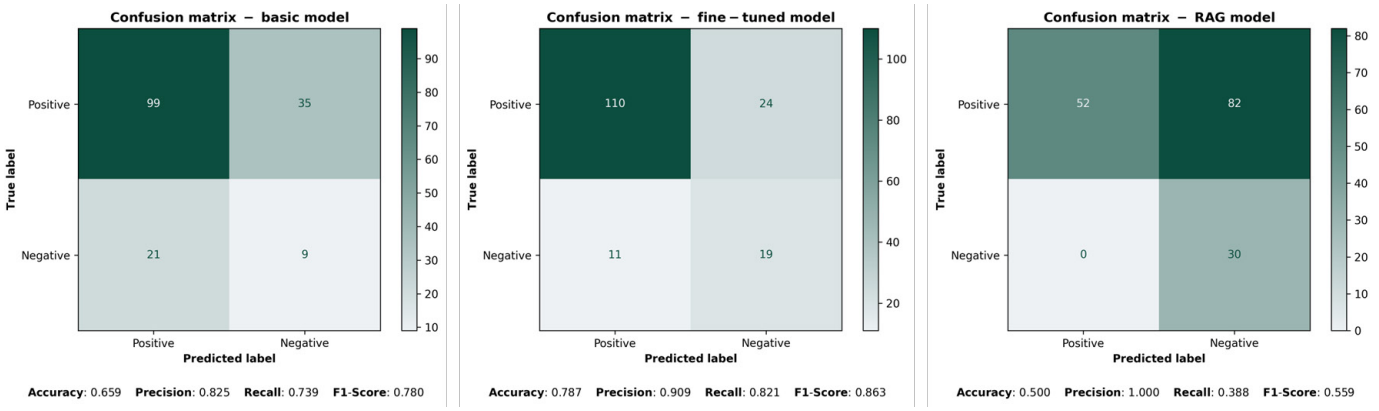


Figure 3: Confusion matrices and evaluation metrics for basic model, fine-tuned model and RAG setup

analyzing satisfactory model performance (see Table 1). Confusion matrices are created for all three models to illustrate the results, displaying the distribution of predictions and enabling the identification of differences in performance among them. These are illustrated in Figure 3, which provides a visual representation of the performance metrics for all three models.

For example, the fine-tuned model achieved a precision of 0.909 and an F1-score of 0.863, indicating its strong performance in generating accurate LCI-related responses. It correctly identified 110 true positives (TP), demonstrating its ability to accurately extract relevant LCI information, such as material quantities and energy consumption associated with lithium production. However, it also recorded 24 false negatives (FN), indicating instances where the model failed to identify relevant LCI data. By contrast, the original model recorded 99 TP, 35 FN, 21 false positives (FP) and 9 true negatives (TN), giving an accuracy rate of 65.9 %. While the model was able to generate some correct responses relating to LCI information, there is significant room for improvement, particularly in reducing the number of FN and improving the overall accuracy of capturing critical data. The RAG model, on the other hand, yielded 52 TP, 82 FN, zero FP and 30 TN, achieving an accuracy of 0.500. Although it demonstrated perfect precision by not generating any FP, its recall was low at 0.388, indicating a significant number of relevant LCI data responses were missed. This suggests that, although the RAG approach can effectively filter out irrelevant data, it may struggle to capture all the LCI information necessary for comprehensive assessments.

5. Case study on SIB documentation

LIBs are already established and widely used, whereas SIBs are still considered as an emerging technology with a comparatively low TRL. For such technologies, prospective assessment is required to identify sustainability potentials and challenges at an early stage of development. Notably, documentation on LIBs tend to be technical and on industrial scale, while SIBs are covered mainly in preliminary studies and on laboratory scale. This highlights the differing data availability between the two technologies. In order to examine the applicability and transferability of the proposed methodology, it is applied to the work of Mazzafarpour et al. (2022) [31], which performed a comparative cradle-to-gate LCA of sodium cathode material. This study is chosen because it provides a detailed LCI and compares multiple synthesis routes, thereby offering a suitable basis to test the methodological transferability. The transfer of the method focusses solely on the LCI section of the study and the method described in section 3 is applied. The results show that the fine-tuned model achieves an accuracy of 0.625 and an F1-score of 0.760, showing balanced precision (0.704) and recall (0.826). The corresponding confusion matrix indicates that most relevant data were correctly identified, with only moderate misclassifications. In contrast, the RAG exhibited very weak performance, with accuracy reduced to 0.219 and an F1-score of 0.242. Both recall (0.174) and precision (0.400) were considerably lower, with the confusion matrix showing systematic misclassification of positive instances as negatives. These results confirm that fine-tuning is more suitable for extracting structured information from the LCI section, whereas RAG is not yet reliable in this context.

6. Discussion

The results reveal distinct performance profiles for the three configurations evaluated in the context of automating the extraction of LCI data from battery related documentation. The basic model shows moderate precision and recall, indicating that while a generic LLM can retrieve some relevant content, it lacks the domain-specific adaption for consistent LCI extraction. The fine-tuned model outperforms the basic model across nearly all metrics, with precision, recall and the F1-score exceeding target values. This suggests that fine-tuning strengthens domain understanding confidence in predictions, although residual ambiguity in the input data and heterogeneous terminology still limit accuracy. The RAG model achieves perfect precision by restricting output to cases with high contextual similarity. However, recall is consistently low. While this conservative output behavior reduces hallucinations and improves traceability, it also omits valid information, which is critical in LCI extraction scenarios where completeness is essential.

The main limitation and challenges are heterogenous input formats, which hinder training and retrieval, RAG's reliance on complete chunking and indexing, and the large, high-quality datasets required for fine-tuning. Furthermore, fixed metric thresholds also only partially reflect extraction quality. In terms of potentials, fine-tuning improves domain understanding and recall, while RAG delivers traceable, high-confidence outputs. Overall, fine-tuned models demonstrate robust and transferable performance across battery chemistries and document types. In contrast, RAG exhibits consistent deficiencies in technical reports and environmental assessments. This highlights the importance of adapting models when using LLMs to evaluate the

environmental impact of new technologies, such as SIBs.

7. Conclusion

This study compared QLoRA-based fine-tuning and RAG approaches for the automated extraction of LCI data from technical documentation related to batteries. Both approaches demonstrate their potential to reduce manual effort required for LCI data collection. A consistent trend was observed across both case studies: fine-tuning produced robust results for both the LIB and SIB case study, while RAG consistently underperforms despite achieving high levels of precision. The fine-tuned model achieved higher recall and F1-scores, indicating its ability to capture a broader range of relevant information. In contrast, RAG delivers higher precision and strong traceability by grounding outputs in retrieved source content. These complementary strengths suggest that fine-tuning is better suited to domain adaptation and coverage, while RAG is more effective in ensuring accuracy and verifiability. However, the observed limitations, such as heterogenous data formats, RAG's dependency on retrieval completeness, and the data requirements for fine-tuning, underline the need for further optimization. A hybrid strategy integrating fine-tuning's adaptability with RAG's retrieval-based validation could address these limitations, enabling the scalable, consistent and transparent acquisition of LCI data from diverse technical sources. This would support more efficient and reliable LCAs in the energy storage sector, and could be extended to other industrial domains with complex documentation. The work therefore is an essential stepstone to accurate, reliable and efficient AI supported LCAs.

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